summing charges and dividing ten, without explanation scores 1/2

			Total		**[3]**
2	**(a)**		<u>mean</u> (value of the) <u>square</u>	M1	
			of the speeds (velocities) of the atoms/particles/molecules	A1	**[2]**
	(b)	**(i)**	$p = \frac{1}{3} \rho < c^2 >$	C1	
			$<c^2> = 3 \times 2 \times 10^5/2.4 = 2.5 \times 10^5$	C1	
			r.m.s speed = 500 ms^{-1}	A1	**[3]**
		(ii)	new $<c^2> = 1.0 \times 10^6$ or $<c^2>$ increases by factor of 4	C1	
			$<c^2> \propto T$ or $3/2\ kT = 1/2\ m<c^2>$	C1	
			$T = \{(1.0 \times 10^6) / (2.5 \times 10^5)\} \times 300$		
			$= 1200$ K	A1	**[3]**
			Total		**[8]**
3	**(a)**	**(i)**	(force) = $GM_1M_2/(R_1 + R_2)^2$	B1	
		(ii)	(force) = $M_1R_1\omega^2$ or $M_2R_2\omega^2$	B1	**[2]**